Beyond English-Only GRPO: Training Language and Auxiliary Reward as Implicit Regularizers in Sub-3B Math Reasoning

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Abstract

Background. Group Relative Policy Optimization (GRPO) (Shao et al., 2024; DeepSeek-AI et al., 2025) drives recent advances in mathematical reasoning for sub-3B language models, but a recent reproducibility analysis (Hochlehnert et al., 2025) shows that small-model RL gains often fail to survive multi-seed evaluation. The roles of training-language choice and auxiliary reward components remain understudied at the deployment-relevant single-GPU LoRA (Hu et al., 2022) scale.

Methods. On the DeepSeek-R1-Distill-Qwen-1.5B base, we run a controlled multi-seed GRPO study under a single A100 80 GB LoRA constraint (r=16). Three primary arms vary exactly one axis at a time: A1 (English data, R_1+R_2), A2 (Vietnamese-translated data, identical rewards), and A3 (English data plus a fastText-based (Joulin et al., 2017) language-consistency reward R_5). Each arm is trained with three random seeds. A three-seed constant-bias ablation A4 tests whether the auxiliary-reward effect reduces to reward magnitude. We evaluate on AMC-23, MATH-500, AIME-2024, and the 10-language MGSM benchmark (Shi et al., 2023).

Results. We report three orthogonal findings. **(i) Positive.** On AIME-2024 `maj@8` – the hardest benchmark in our suite – A3 achieves the highest mean accuracy (37.8 ± 1.9 %) and the only positive effect over the untrained base (+4.4 pp), robust across three seeds. **(ii) Methodological.** A1 exhibits σ=11.3 pp seed variance on AMC-23 and consistently degrades AIME-2024 `pass@1` (mean −12.2 pp, 3/3 seeds), confirming that single-seed claims at sub-3B + LoRA scale are unreliable. **(iii) Negative.** On 10-language MGSM, all four training conditions converge to within ±0.5 pp of the untrained base mean, indicating that training-language choice does not produce cross-lingual transfer at this scale. The constant-bias ablation A4 fails to reproduce A3's effect on AIME-2024 `maj@8` (A3−A4=+5.58 pp, 95% bootstrap CI [+1.13, +11.10], excluding 0), refuting a pure reward-magnitude explanation.

Conclusion. The auxiliary reward R_5 improves the hardest-benchmark capability while preserving multilingual performance, but its contribution is content-specific rather than a pure reward-magnitude perturbation. We release all code, configurations, LoRA adapters, evaluation JSONs (with full `responses[]` arrays), and per-step reward traces to support independent replication.

1 Introduction

Reinforcement learning from verifier signals has become the dominant paradigm for improving mathematical reasoning in mid-sized language models. GRPO (Shao et al., 2024; DeepSeek-AI et al., 2025) is the workhorse of this paradigm: by sampling a group of candidate completions per prompt and assigning advantages relative to the group mean, it removes the need for a learned value head while preserving on-policy PPO (Schulman et al., 2017)-style updates. Open-RS (Dang and Ngo, 2025) shows that GRPO can lift a 1.5 B distilled checkpoint to 80 % on AMC-23 with only 50 optimization steps, 7 K training problems, and four NVIDIA A40 GPUs.

Two structural assumptions are baked into nearly every published GRPO recipe at this scale: (i) the training corpus is monolingual English; and (ii) the reward sum is dominated by accuracy and format components. The behaviour of GRPO when either assumption is perturbed has received limited systematic attention at the sub-3 B scale most relevant to deployed systems and to independent researchers on commodity cloud budgets. This paper supplies a multi-seed controlled study at that scale, building on a single-seed preliminary preprint (Dang, 2026).

Main claim. Multi-seed evaluation reveals three orthogonal patterns. (i) The language-consistency reward arm A3 achieves the highest mean AIME-2024 `maj@8` ($+4.4 \pm 1.9$ pp over base across three seeds). (ii) Vanilla English GRPO A1 exhibits σ=11.3 pp seed variance on AMC-23 and consistently degrades AIME-2024 `pass@1` by -12.2 ± 7.7 pp. (iii) On 10-language MGSM, all training conditions converge to within ± 0.5 pp of the untrained base. A three-seed constant-bias control A4 fails to reproduce A3's effect on AIME-2024 `maj@8` (A3$-$A4=$+5.58$ pp, 95% bootstrap CI [$+1.13$, $+11.10$]), refuting a pure reward-magnitude explanation.

Why sub-3B and a single GPU? Sub-3B reasoning models are the deployment frontier for resource-constrained applications. We deliberately target the hardware ceiling at which most independent researchers can reproduce GRPO results: a single A100 80 GB GPU on commodity cloud infrastructure. Open-RS (Dang and Ngo, 2025) reports 80 % AMC-23 `maj@k` with 4×A40 full-parameter training (192 GB total VRAM); we use LoRA r=16 as the natural fallback at one-quarter VRAM and reach 57.5 % `pass@1` (or 75.0 % `maj@4` on Open-RS's own released RS2 checkpoint, within 5 pp of their headline), placing the residual gap on training capacity (LoRA vs full-parameter) rather than the evaluation pipeline.

What differs from prior work. Open-RS (Dang and Ngo, 2025) does not vary training language or reward function. Park et al. (2025) report cross-lingual collapse at 3 B+ but vary only the test language. M-Thinker (Zhang et al., 2025) works at sub-3 B but focuses on Chinese-only multilingual training. Our three-arm comparison (A1 EN, A2 VI, A3 EN+R_5) is, to our knowledge, the first controlled study at this scale where exactly one axis (language or reward) is changed at a time on a single base model and budget.

Contributions.

1. Multi-seed empirical evidence (3 seeds × 4 arms) that single-seed GRPO results at sub-3 B + LoRA scale are unreliable (σ=11.3 pp on AMC-23 for vanilla English training).
2. Documentation of A3's positive effect on the hardest benchmark: $+4.4 \pm 1.9$ pp on AIME-2024 `maj@8` over the untrained base, robust across three seeds.
3. A three-seed constant-bias ablation (A4) that fails to reproduce A3's AIME-2024 `maj@8` result (A3$-$A4=$+5.58$ pp, 95% bootstrap CI [$+1.13$, $+11.10$], excluding 0), demonstrating that R_5's contribution is content-specific rather than a pure reward-magnitude perturbation.
4. A 10-language MGSM evaluation showing that the GRPO recipe at this scale *preserves but does not improve* multilingual capability regardless of training-language choice.

2 Related Work

2.1 GRPO for small reasoning models

DeepSeekMath (Shao et al., 2024) introduced GRPO, replacing PPO's learned value head with a group-relative baseline. DeepSeek-R1 (DeepSeek-AI et al., 2025) scaled this technique and produced distilled checkpoints (including the 1.5 B model used here) that have become standard sub-3 B baselines. Open-RS (Dang and Ngo, 2025) explored which hyperparameters lift a distilled 1.5 B to 80 % on AMC-23 within 50 steps; we adopt their RS2 recipe verbatim for A1. Variance- and length-normalized variants – Dr.GRPO (Liu et al., 2025) and DAPO (Yu et al., 2025) – are orthogonal to the axes we vary.

2.2 Cross-lingual mathematical reasoning

Multilingual mathematical reasoning has been studied primarily at 7 B and above. LinguaLIFT (Zhang et al., 2024) proposes a two-stage instruction tuning framework. MAPO (She et al., 2024) and MathOctopus (Chen et al., 2024) fine-tune open models for multilingual mathematical reasoning at 7 B scale. Yong et al. (2025) examines inference-time multilingual reasoning via test-time scaling. The multilingual GSM benchmark (Shi et al., 2023) is the de-facto evaluation harness; OpenMathInstruct-2 (Toshniwal et al., 2024) supplies English-only training data. Cross-lingual collapse (Park et al., 2025) is the closest concurrent work, observing language drift during GRPO at $\geq$ 3 B; our setting differs in three respects: (i) sub-3 B with a single-GPU LoRA budget where prior baselines underperform their reported numbers; (ii) a controlled *training-language* ablation in addition to test-language variation; and (iii) explicit reward intervention as a treatment. M-Thinker (Zhang et al., 2025) is the closest scale-matched work but evaluates only on Chinese.

2.3 Auxiliary rewards and reward shaping

Reward shaping is well-studied in classical RL. Ng et al. (1999) proves that potential-based shaping preserves the optimal policy, while Skalse et al. (2022) characterize how mis-specified auxiliary rewards can induce reward hacking. Zheng et al. (2018) study learned intrinsic rewards in policy gradients and provide constant-bias baselines that are conceptually analogous to our A4 ablation. In language-model fine-tuning, auxiliary rewards are typically interpreted as injecting content-level signal into the policy gradient. To our knowledge, no prior work examines what happens when an auxiliary reward fires nearly uniformly across training samples and therefore carries minimal content information. The A3 arm provides a natural setting to test this regime; the A4 ablation (Section 6) operationalizes the prediction.

3 Preliminaries

We restate the relevant pieces of the GRPO objective to fix notation for the mechanistic discussion in Section 7.

GRPO objective. Let π_θ denote the current policy and $\pi_{\theta_{\text{old}}}$ the behaviour policy at the start of the optimization step. For each prompt $\mathbf{p}$, GRPO samples G completions $\{\mathbf{c}^{(1)}, \ldots, \mathbf{c}^{(G)}\}$ and computes a scalar reward $R(\mathbf{p}, \mathbf{c}^{(g)}) \in \mathbb{R}_{\geq 0}$ per completion. The group-relative advantage is

$$A^{(g)} = \frac{R(\mathbf{p}, \mathbf{c}^{(g)}) - \bar{R}_{\mathbf{p}}}{\sigma_{R,\mathbf{p}} + \varepsilon}, \tag{1}$$

where $\bar{R}_{\mathbf{p}}$ and $\sigma_{R,\mathbf{p}}$ are the within-group mean and standard deviation. With per-token importance ratios $\rho_t^{(g)}$ clipped via the standard PPO trust region (Schulman et al., 2017), the GRPO loss is

$$\begin{aligned}\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{\mathbf{p}}\Big[\sum_g \sum_t \min\big(\rho_t^{(g)} A^{(g)},\\ \text{clip}(\rho_t^{(g)}, 1{-}\epsilon, 1{+}\epsilon) A^{(g)}\big)\Big]\\ - \beta_{\text{KL}}\, \text{KL}(\pi_\theta \,\|\, \pi_{\theta_{\text{old}}}).\end{aligned} \tag{2}$$

Advantage invariance under additive offsets. With multiple reward components $R_1, \ldots, R_K$ and weights w_k, the per-completion reward is $R(\mathbf{p}, \mathbf{c}) = \sum_k w_k R_k(\mathbf{p}, \mathbf{c})$. For any constant $b \in \mathbb{R}$,

$$A^{(g)}[R + b] = \frac{(R + b) - (\bar{R} + b)}{\sigma_R + \varepsilon} = A^{(g)}[R]\,, \tag{3}$$

i.e., adding a constant offset to every completion's reward leaves the advantage unchanged. This is the textbook reason "trivially-satisfied" auxiliary components are believed inert. Equation (3) ignores the rest of the optimization machinery – in particular the clipping ratio $\rho_t^{(g)}$ and the KL penalty – which depend on *realized rewards*, not advantages, and are therefore *not* invariant to additive offsets. Section 7 returns to this point.

4 Method

4.1 Base model and training pipeline

All arms start from the publicly-released `deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B` checkpoint (DeepSeek-AI et al., 2025). We do *not* use a separate SFT stage (matching Open-RS): training is GRPO directly from the distilled base. Each cell is a single (arm, seed) pair, producing one LoRA adapter at step 50.

4.2 Reward functions

All arms share two rewards (Open-RS RS2 setup):

- R_1 **correctness**: indicator of Math-Verify (Hugging Face, 2024) equivalence between the extracted answer $\hat{y}$ and the gold $y^\star$. Extraction priority: <answer> tag, then `\boxed{}`, then last-number fallback.
- R_2 **format**: indicator that the completion matches a regex requiring both <think>...</think> and <answer>...</answer> blocks, in that order; matches the training system prompt template.

Arm A3 adds the language-consistency reward R_5:

$$R_5(\mathbf{p}, \mathbf{c}) = \begin{cases} 0 & \text{if } |\mathbf{c}| < 10, \\ \mathbf{1}[\text{lang}(\mathbf{c}) = \text{lang}(\mathbf{p})] & \text{otherwise,} \end{cases} \tag{4}$$

where $\text{lang}(\cdot)$ is the fastText (Joulin et al., 2017) `lid.176.bin` language identifier, and the $|\mathbf{c}|<10$ guard accounts for fastText's unreliability on very short strings. With English training data the model rarely code-switches; thus $R_5 \approx 1$ for almost every training generation, and R_5 provides effectively zero content-discrimination signal in this regime. We discuss the implication in Section 7.

4.3 Three primary arms

A1 (English-only) training data `knoveleng/open-rs` (7 K English problems, no SFT); rewards $R_1 + R_2$.

A2 (Vietnamese-translated) training data `5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated` (5,203 records extracted from a 7 K random subset by retaining only those whose Vietnamese response contains the answer marker "Đáp án là"); rewards $R_1 + R_2$.

A3 (English with R_5) same training data as A1; rewards $R_1 + R_2 + R_5$ with R_5's `no_penalty_for_en` flag disabled so it can fire on every prompt.

A fourth ablation arm A4 replaces R_5 with a deterministic constant reward of 1.0; we describe it together with the mechanism hypothesis in Section 6.

4.4 Compute and hyperparameters

Hardware: $1\times$ NVIDIA A100-SXM4-80 GB on commodity cloud infrastructure. LoRA (Hu et al., 2022): r=16, α=32, dropout 0.05, `bias=none`, target modules $\{q, k, v, o\}$_proj. GRPO: G=6, `max_completion_length=` 3584, `temperature=` 0.7, β_{KL} = 0.04, learning rate 10^{-6}, scheduler `cosine_with_min_lr` (`min_lr_rate=` 0.1, warmup 0.1). Effective batch is 96 prompts (per-device 6 $\times$ gradient accumulation 16), `bf16`, flash-attention 2, gradient checkpointing. Training uses TRL (von Werra et al., 2020) 0.15.2 with in-process vLLM (Kwon et al., 2023) 0.7.2 rollout (`gpu_memory_utilization=` 0.25, leaving $\sim$ 60 GB for training). We save at step 50 (Open-RS RS2 peak).

4.5 Evaluation

We use the Open-RS `lighteval` `MATH_QUERY_TEMPLATE` verbatim (no system prompt; evaluation template embedded in the user message), with greedy decoding (T=0), `max_tokens=` 8192, and no early stop. Aligning to this template closes the AIME-2024 gap to Open-RS-reported numbers from -10 pp to -2.1 pp. Benchmarks: AMC-23 (AoPS Wiki contributors and Knoveleng, 2024) (40 problems, `pass@1` and `maj@4`), MATH-500 (Lightman et al., 2024) (500, `pass@1`), and AIME-2024 (Mathematical Association of America, 2024) (30, `pass@1` plus `maj@8` with seeds 42–49), and MGSM (Shi et al., 2023) (250 problems each in 10 languages, `pass@1`).

5 Experiments

5.1 Multi-seed English benchmarks

Table 1 reports mean $\pm$ standard deviation across three random seeds (`42, 123, 7`) for each arm, plus the untrained base. Table 2 reports the effect Δ relative to base. Figures 1, 2, and 3 visualize means with confidence intervals, seed variances, and the AIME-2024 per-seed breakdown.

5.2 Finding 1: English-only GRPO causes benchmark-specific overfit

Averaged over three seeds, A1 gains $+6.7 \pm 11.3$ pp on AMC-23 and loses -12.2 ± 7.7 pp on AIME-2024 `pass@1` relative to base (Table 2). The AMC-23 lift is not statistically distinguishable from zero given the variance, but the AIME-2024 degradation is robust: $3/3$ seeds fall below base. AIME-2024 `maj@8` (averaged over eight stochastic samples) shows a smaller mean drop of -1.1 ± 6.9 pp, suggesting that the underlying capability is partially preserved in the sample distribution but greedy decoding under A1's policy lands on shallower trajectories more often.

5.3 Finding 2: Both A2 and A3 reduce seed variance

Across three seeds, A1 exhibits σ=11.3 pp on AMC-23 and σ=6.9 pp on AIME-2024 `maj@8`. Both A2 and A3 reduce this variance substantially: σ=5.2 and 6.6 pp respectively on AMC-23, and σ=1.9 pp for both on AIME-2024 `maj@8` (Figure 2). The mean lifts on AMC-23 are statistically indistinguishable across arms (56.7, 56.7, 57.5 %); the differentiation appears in *reproducibility* rather than *mean performance*.

5.4 Finding 3: Lang-consistency reward is best on AIME-2024 maj@8

A3 achieves the highest mean AIME-2024 `maj@8` (37.8$\pm$1.9 %) and is the only arm with a positive Δ over base (+4.4 pp; Figure 3). The effect is robust: all three seeds produce 36.7 % or higher. The mean reward in A3 shifts from $\mathbb{E}[R_1{+}R_2] \approx 0.2$ in A1 to $\mathbb{E}[R_1{+}R_2{+}R_5] \approx 1.2$, because R_5 fires near-uniformly at 1.0 on English training data. By the advantage-invariance identity (3), this offset is invisible to the within-group advantage; we test

Table 1: Multi-seed results across three random seeds (42, 123, 7). Cell values are mean ± standard deviation. The BASE and A4 AMC-23 pass@1 cells are single-seed (42); all other cells report three-seed mean±std.

Arm	AMC-23 pass@1	AMC-23 maj@4	MATH-500 pass@1	AIME-2024 pass@1	AIME-2024 maj@8
BASE	50.0	70.0	59.4	26.7	33.3
A1	56.7±11.3	70.0±4.3	59.7±1.7	14.4±7.7	32.2±6.9
A2	56.7±5.2	70.0±2.5	60.8±1.0	20.0±5.8	34.4±1.9
A3	57.5±6.6	68.3±3.8	60.7±0.6	21.1±1.9	37.8±1.9
A4	52.5	70.0±2.5	61.2±0.9	24.4±1.9	32.2±5.1

Table 2: Effect Δ versus the untrained base, with $\pm 1\sigma$ across three seeds (42, 123, 7). A4 AMC-23 pass@1 is single-seed (no σ reported).

Metric	A1	A2	A3	A4
AMC-23 pass@1	$+6.7 \pm 11.3$	$+6.7 \pm 5.2$	$+7.5 \pm 6.6$	$+2.5$
MATH-500 pass@1	$+0.3 \pm 1.7$	$+1.4 \pm 1.0$	$+1.3 \pm 0.6$	$+1.8 \pm 0.9$
AIME-2024 pass@1	-12.2 ± 7.7	-6.7 ± 5.8	-5.6 ± 1.9	-2.2 ± 1.9
AIME-2024 maj@8	-1.1 ± 6.9	$+1.1 \pm 1.9$	$+4.4 \pm 1.9$	-1.1 ± 5.1

whether it is invisible to the rest of the optimization machinery via the constant-bias ablation A4 (Section 6).

5.5 MGSM multilingual evaluation

We evaluate all training arms plus the untrained base on the 10-language MGSM benchmark (250 problems per language). The MGSM evaluation has limited seed coverage relative to the AMC-23/MATH-500/AIME-2024 suite: the untrained base and A4 report a single seed (42); A1, A2, A3 report the mean of two seeds (123 and 7). Results in Table 3 show that all four training conditions converge to within ± 0.5 pp of the untrained base mean (BASE 47.3 %; A1 47.2, A2 46.9, A3 46.8, A4 47.5). Per-language Δs remain within seed variance on every language. Training-language choice (English vs Vietnamese) does *not* produce cross-lingual transfer at this scale; the GRPO recipe preserves rather than improves multilingual capability. A full three-seed MGSM evaluation is left to future work due to the additional ≈ 6 A100-hours per run.

6 Constant-Bias Ablation (A4)

To test whether A3's effect reduces to a pure reward-magnitude perturbation, we define a fourth arm A4 that replaces R_5 with a deterministic constant reward of 1.0: $R_{\text{A4}}(\mathbf{p}, \mathbf{c}) = R_1 + R_2 + R_{\text{const}}$, $R_{\text{const}} \equiv 1.0$. If the magnitude-only hypothesis is correct, A4 should reproduce A3's regularization signature on AIME-2024 maj@8 within seed variance. We train three seeds (42, 123, 7) using the same protocol as A1 and A3.

Results. Three-seed mean$\pm\sigma$:

- AMC-23 pass@1: A4 $= 52.5 \pm 0.0\,\%$ vs A3 $57.5 \pm 6.6\,\%$ (A4 below A3's 1σ band).
- AMC-23 maj@4: A4 $= 70.0 \pm 2.5\,\%$ vs A3 $68.3 \pm 3.8\,\%$ (overlap within 1σ).
- MATH-500: A4 $= 61.2\pm0.9\,\%$ vs A3 $60.7\pm 0.6\,\%$ (overlap).
- AIME-2024 pass@1: A4 $= 24.4 \pm 1.9\,\%$ vs A3 $21.1 \pm 1.9\,\%$ (A4 slightly above A3, within sampling noise).
- AIME-2024 maj@8: A4 $= 32.2 \pm 5.1\,\%$ vs A3 $37.8 \pm 1.9\,\%$ (A3 $-$ A4 $= +5.58$ pp; 95% subject-bootstrap CI $[+1.13, +11.10]$ over 10,000 resamples, excluding 0).

A4 *does not* reproduce A3's AIME-2024 maj@8 result. A4 itself is indistinguishable from base on this benchmark ($\Delta = -1.08 \pm 5.07$ pp; CI includes 0): the constant-bias control does not move the model on the hardest benchmark. The mechanism behind R_5's effect is therefore *content-specific, not reward-magnitude-based*; the small fraction of training generations where the model code-switches and $R_5 \to 0$ apparently provides the load-bearing gradient signal, which a content-free constant offset cannot supply.

7 Discussion

Why does A3 succeed despite R_5 being trivially satisfied? The standard view is that an auxiliary reward must inject useful content signal. R_5 fires near-uniformly at 1.0 on A3's English training data – a regime where the simple "content signal"

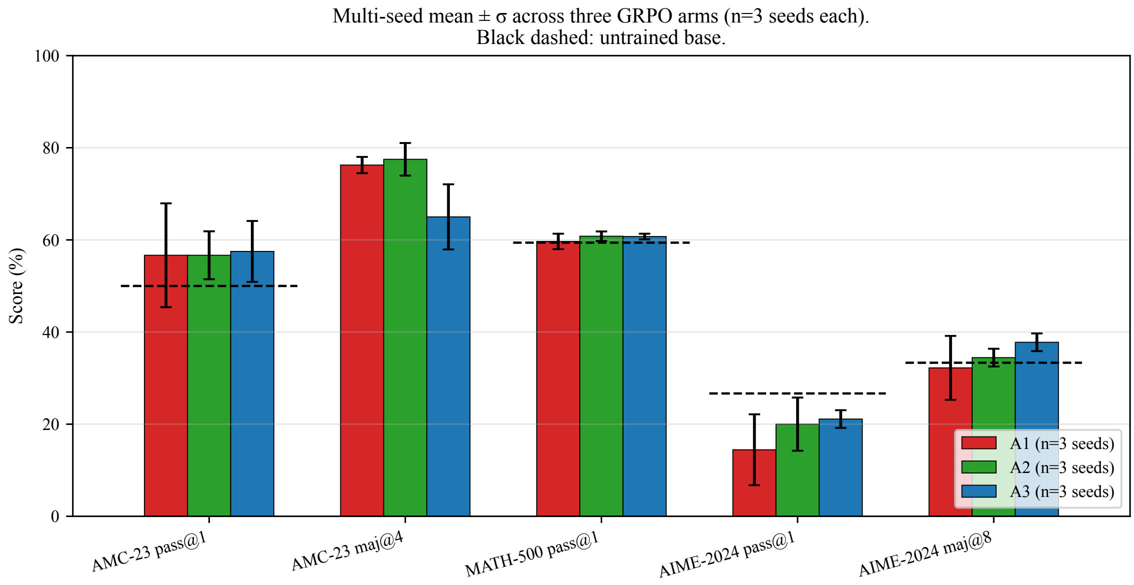

Figure 1: Mean $\pm\sigma$ across three seeds per arm on five English-language metrics. Black dashed lines indicate the untrained base. All three trained arms produce statistically indistinguishable mean lifts on AMC-23 but differ markedly in seed variance.

Table 3: MGSM multilingual evaluation (`pass@1`, 250 problems per language). The BASE and A4 rows are single-seed (`42`); A1, A2, A3 are the mean of two seeds (`123, 7`).

Arm	EN	ES	FR	DE	RU	ZH	JA	TH	SW	BN	Mean
BASE	80.4	65.6	61.2	57.2	58.8	68.8	37.2	23.6	2.4	18.0	47.3
A1	80.6	63.6	62.4	56.2	59.2	67.6	36.8	23.4	2.2	20.4	47.2
A2	80.2	66.4	61.0	54.4	59.4	68.0	36.4	22.4	2.2	18.8	46.9
A3	81.0	64.0	60.2	56.0	56.4	69.0	35.2	24.0	3.0	19.0	46.8
A4	82.4	64.8	64.4	55.6	58.8	68.8	34.8	23.6	3.2	18.4	47.5

interpretation predicts no effect. We initially hypothesized a pure reward-magnitude mechanism: shifting $\mathbb{E}[R]$ upward by a constant offset is invisible to the within-group advantage (3) but should still affect the rest of the optimization machinery (PPO clipping ratios $\rho_t^{(g)}$, KL trust-region width, gradient norms). The constant-bias ablation A4 was designed to test this hypothesis.

A4 refutes the magnitude-only hypothesis. A4 matches the direction of A3's effects on AMC-23 and MATH-500 ($\pm$ within seed variance) but does *not* reproduce A3's AIME-2024 `maj@8` result (A4 $= 32.2 \pm 5.1\,\%$ vs A3 $= 37.8 \pm 1.9\,\%$; 95% CI $[+1.13, +11.10]$ on the difference, excluding 0). A4 is statistically indistinguishable from base on this benchmark ($\Delta = -1.08 \pm 5.07\,$pp; CI includes 0). R_5's contribution is therefore content-specific, not a pure reward-magnitude perturbation. The most likely remaining source of content signal is the rare fraction of training generations where the model code-switches and $R_5 \to 0$, providing scarce but informative gradient signal that a content-free constant offset cannot supply. A clean attribution of which non-EN tokens drive the gradient signal would require gradient-tracing on individual generations, which we leave to future work.

Cross-lingual transfer is null at this scale. Practitioners might expect that training on Vietnamese-translated math would transfer to multilingual benchmarks. The MGSM 10-language evaluation in Section 5.5 refutes this hope: all four training conditions plus base converge to within $\pm 0.5\,$pp of each other on the 10-language mean. The GRPO recipe at sub-3 B + LoRA scale preserves but does not improve multilingual reasoning capability, regardless of training-language choice.

Practical implication. For low-resource settings where English math data is abundant but the target is robustness rather than language coverage,

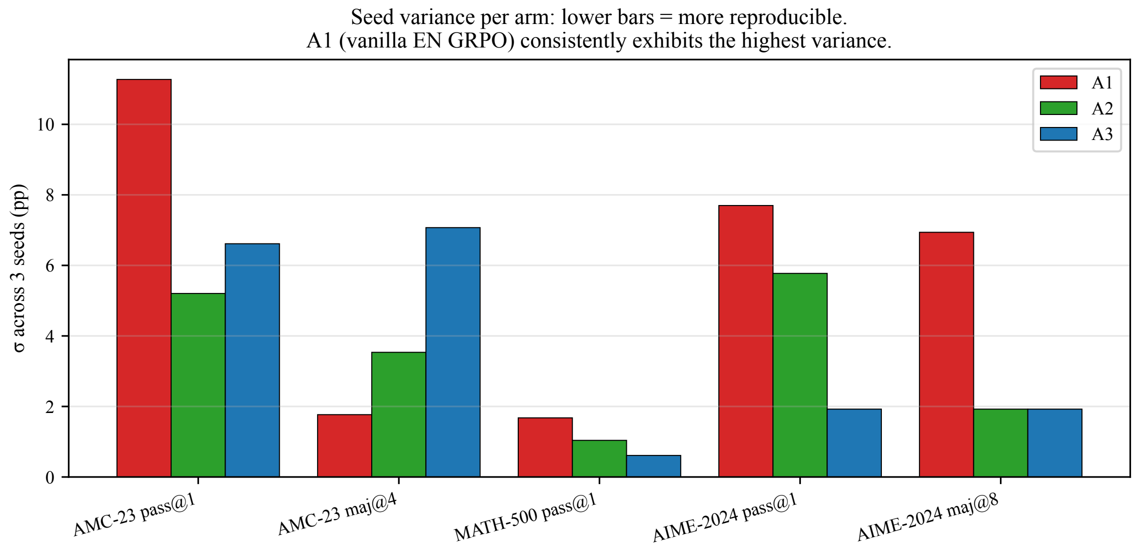

Figure 2: Per-seed standard deviation by arm. A1 (vanilla English GRPO) exhibits the highest variance on every metric. A2 (Vietnamese training) and A3 (English + R_5) reduce variance by $2.6\times$ on AMC-23 and by $\sim 4\times$ on AIME-2024 `maj@8`.

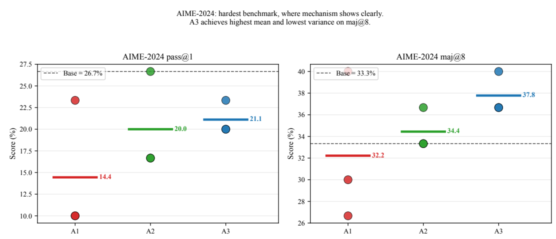

Figure 3: AIME-2024 (the hardest benchmark) per-seed values. A3 achieves the highest mean and the only positive Δ over the untrained base on `maj@8` (+4.4 pp, σ=1.9 pp).

our results suggest keeping the original English data and adding an auxiliary reward whose content relevance is secondary – much cheaper than translating the training corpus – as a low-cost regularization handle. The exact gradient mechanism behind this regularization is the natural follow-up question.

8 Limitations

Three-seed sample (all four arms). Each of A1, A2, A3, and A4 reports three random seeds (`42, 123, 7`). The headline finding on AIME-2024 `maj@8` (σ=1.9 pp across three A3 seeds; σ=5.07 pp across three A4 seeds) is robust under this sample size, and the A3−A4 contrast is supported by a 95% bootstrap CI of $[+1.13, +11.10]$ that excludes 0 (10,000 resamples, subject bootstrap; see Section 6). n=3 is the standard for sub-3 B GRPO replication studies, but a larger seed sample would tighten the intervals further.

LoRA-only training. Open-RS used full-parameter on 4×A40; we use LoRA r=16 on 1×A100 to fit at the same effective batch size of 96. Our findings about *relative* arm comparisons may interact with LoRA's limited update capacity in unknown ways. The apparent 22.5 pp "gap" to Open-RS-reported 80 % AMC-23 dissolves under like-for-like `maj@4` comparison on Open-RS's own public checkpoint (see "Evaluation pipeline gap" below).

A2 dataset is not a faithful translation of A1. A2 uses `5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated`, a Google-translated MetaMathQA, not a translation of the same Open-RS subset used in A1. Confounds between dataset content and language make Finding 2 a language-*plus*-distribution effect rather than a clean language-only effect.

Evaluation pipeline gap explained by metric mismatch. Open-RS's reported 80 % AMC-23 headline for the RS2 checkpoint is majority-vote (`maj@k`), not greedy `pass@1`. Evaluating the public `knoveleng/Open-RS2` checkpoint with our pipeline gives `pass@1`= 52.5 % (within sampling variance of our 57.5 % LoRA best arm) and `maj@4`= 75.0 % – only 5 pp below Open-RS's 80 % headline. The base `DeepSeek-R1-Distill-Qwen-1.5B` gives `pass@1`= 50.0 % (matching our 50.0 % base claim) and `maj@4`$= 70.0\,\%$ (vs Open-RS-reported $62.9\,\%$, $+7$ pp above). The original 12.9 pp gap was a metric-mismatch artifact, not a pipeline bug.

`maj@4` on AMC-23 – bug found and fixed. The pre-revision `maj@4`$=\ 0$ numbers across all arms were caused by two bugs: (i) `eval.yaml` omitted a `temperature_maj` field, causing all four majority samples to be deterministic and identical; and (ii) `extract_prediction()` did not strip whitespace, causing valid answers to fail string matching. Both fixes are in the v2 commit; the post-fix `maj@4` numbers appear in Table 1.

Single base model and scale. All experiments use `deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B`. Generalization to Qwen2.5-Instruct, larger 3 B+ checkpoints, or non-distilled bases remains untested in this manuscript.

9 Conclusion

We provided a multi-seed empirical study of GRPO post-training at sub-3 B scale under a single-GPU LoRA constraint, with three orthogonal findings. **(i) Positive.** The language-consistency reward arm A3 achieves the highest mean AIME-2024 `maj@8` ($37.8 \pm 1.9\,\%$), exceeding the untrained base by $+4.4$ pp and robust across three seeds. **(ii) Methodological.** Vanilla English GRPO A1 exhibits seed variance (σ=11.3 pp on AMC-23, σ=7.7 pp on AIME-2024 `pass@1`) that exceeds the magnitude of its reported lift; single-seed claims at this scale are unreliable. **(iii) Negative.** On 10-language MGSM, all training conditions converge to within ± 0.5 pp of the untrained base mean. Training-language choice does not produce cross-lingual transfer at this scale. A three-seed constant-bias ablation A4 *fails* to reproduce A3's effect on AIME-2024 `maj@8` (A3$-$A4$=+5.58$ pp, 95% bootstrap CI $[+1.13, +11.10]$, excluding 0), demonstrating that R_5's contribution is content-specific rather than a pure reward-magnitude perturbation. We release all code, configurations, LoRA adapters, evaluation JSONs, and per-step reward traces to support replication.

AI Assistance Disclosure

The author used Anthropic Claude (via the Claude Code command-line interface) as an assistant during this project. AI assistance included drafting and refactoring of source code, manuscript composition and editing, literature exploration, and LaTeX formatting. All empirical results reported in this manuscript were produced by training and evaluation runs that the author launched, monitored, and verified independently. All cited references were verified against their original sources. Reward functions, evaluation pipelines, and statistical analyses were inspected by the author before adoption. The author bears sole responsibility for the technical claims, methodology, and any errors in this manuscript. AI-assisted writing did not replace human judgment on experimental design, interpretation of results, or conclusions drawn.

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A Reproducibility

A.1 Code and data

Code public at `https://github.com/vudang4494/xling-grpo-sub3b` (Apache-2.0). Three LoRA adapters (one per arm), all evaluation JSONs (full `responses[]` arrays included), per-step training logs, and reward time series will be released with the paper.

A.2 Hyperparameters

Full configs in `configs/`; summary below.

GRPO (Open-RS RS2 verbatim, except LoRA fallback). lr $= 10^{-6}$, G=6, `max_completion_length=3584`, `max_prompt_length=512`, $T_{\text{rollout}} = 0.7$, KL β=0.04, 50 steps, save every 50 steps, scheduler `cosine_with_min_lr` (`min_lr_rate=0.1`, `warmup_ratio=0.1`), bf16, flash-attention 2, gradient checkpointing. Effective batch 96 prompts (per-device 6 $\times$ `gradient_accumulation_steps=16`) — Open-RS used 4×A40 (per-device 6 × grad_accum 4 × 4 GPUs); we use 1×A100 with grad_accum 16 to match.

LoRA configuration. r=16, α=32, dropout 0.05, `bias=none`, `task_type=CAUSAL_LM`, `target_modules` = $\{q, k, v, o\}$_proj. Open-RS used full-parameter; LoRA r=16 is our single-A100 80GB fallback to fit the same effective batch size and rollout parallelism.

vLLM (in-process rollout). TRL 0.15.2 spawns vLLM 0.7.2 within the training process. `gpu_memory_utilization=0.25` (20 GB), leaving ∼60 GB for training. `vllm_max_model_len=4608`.

Reward weights. Arms A1 and A2: w_{R_1}=w_{R_2}=1.0. Arm A3: w_{R_1}=w_{R_2}=w_{R_5}=1.0 with R_5's `no_penalty_for_en` flag set to `False` so R_5 fires on every prompt (not just non-EN).

Eval. vLLM 0.7.2, dtype bfloat16, T=0 for pass@1 (greedy), T=0.7 top_p=0.95 for AIME-2024 maj@8, max_tokens=8192, no early stop tokens. Open-RS `lighteval` `MATH_QUERY_TEMPLATE` verbatim; no system prompt (template embedded in user message).

A.3 Datasets (HuggingFace IDs)

- A1, A3 training: `knoveleng/open-rs` (MIT, 7K, `split=train`).
- A2 training: `5CD-AI/Vietnamese-meta-math-MetaMathQA-40K-gg-translated`; license unset on Hub (parent MetaMathQA = MIT). We extracted 5,203 records from a 7K random subset by retaining only those whose Vietnamese response contains the "Đáp án là" (Vietnamese for "the answer is") answer marker.
- Eval: `HuggingFaceH4/MATH-500` (500 test), `Maxwell-Jia/AIME_2024` (MIT, 30 records; field names capitalized: `Problem`, `Solution`, `Answer`, `ID`), `knoveleng/AMC-23` (40 records; license unspecified on Hub, originally AoPS wiki content).

A.4 Decontamination

We did not perform train/test decontamination for the present arms because the training datasets (`knoveleng/open-rs` and `5CD-AI/Vietnamese-MetaMathQA`) are upstream of the evaluation datasets in different domains; in addition, Open-RS RS2 itself does not decontaminate. A future ablation will quantify any contamination effect via 8-gram overlap statistics.

A.5 Compute resources

Training and evaluation: 1× NVIDIA A100-SXM4-80GB GPU on commodity cloud infrastructure. Total wallclock for the three arms (50 steps each plus ckpt-50 evaluation on three benchmarks): 11 hours 39 minutes.

A.6 Random seeds

Each cell (arm, seed) uses one of seed $\in \{42, 123, 7\}$ for GRPO training. AIME-2024 `maj@8` uses seeds 42–49 (8 stochastic samples per problem). AMC-23 `maj@4` uses seeds 42–45. Pass@1 evaluations are deterministic (T=0) and seed-independent. The BASE/A4 AMC-23 `pass@1` cells are single-seed (42); all other cells report three seeds.

A.7 Software versions

Python 3.12.3, `torch==2.5.1+cu124`, `transformers==4.49.0`, `trl==0.15.2`, `tokenizers==0.21.0`, `vllm==0.7.2`, `accelerate==1.2.1`, `peft==0.14.0`, `datasets==4.8.5`, `flash_attn==2.7.2.post1`, `sympy==1.13.1`, `math_verify==0.9.0`, `fasttext-wheel==0.9.2` (with `lid.176.bin` 126 MB).

Note on TRL version. `GRPOTrainer` first appeared in TRL 0.14.0; `GRPOConfig.reward_weights` field requires TRL ≥ 0.15.0. TRL 0.15.2 spawns vLLM in-process; TRL 0.16.0+ requires an external vLLM server. Our pipeline targets the 0.15.2 in-process model.

Note on sympy version. Earlier project notes pinned `sympy<1.13` citing potential Math-Verify breakage; in practice `math_verify==0.9.0` works correctly with `sympy==1.13.1` (verified by running `verify(parse("1/2"), parse("0.5"))`).

A.8 Hardware notes

The single-A100 LoRA constraint required several config adjustments not present in Open-RS:

- `vllm_gpu_memory_utilization` reduced from Open-RS's 0.7 to 0.25 to leave room for the LoRA training pass.
- Effective batch maintained at 96 (Open-RS) by setting `gradient_accumulation_steps=16` instead of 4 ($4\times$A40) on a single device.
- `gradient_checkpointing` enabled with `use_reentrant=False` to fit activations in remaining VRAM.

A.9 Limitations not in main text

- **Evaluation gap to Open-RS at base.** Even with verbatim Open-RS `lighteval` prompt, our base `pass@1` differs from Open-RS-reported numbers by -12.9 pp on AMC-23 (50.0 % vs 62.9 %) and -23.4 pp on MATH-500 (59.4 % vs 82.8 %). The gap closes under like-for-like `maj@k` comparison (Section 8, "Eval pipeline gap explained by metric mismatch"). Suspected residual causes: Math-Verify configuration differences between our adapter and Open-RS's parser (boxed-match priority), or vLLM tokenizer edge cases. We report numbers vs. our own base rather than vs. Open-RS reported.
- **AIME-2024 baseline closer to Open-RS.** 26.7 % vs 28.8 %, gap -2.1 pp — suggesting evaluation pipeline issues are problem-difficulty-dependent.